

Topic 13 – Electromagnetic induction

Students should:	Maths skills
13.1P Explain how to produce an electric current by the relative movement of a magnet and a conductor a on a small scale in the laboratory b in the large-scale generation of electrical energy	
13.2 Recall the factors that affect the size and direction of an induced potential difference, and describe how the magnetic field produced opposes the original change	5b
13.3P Explain how electromagnetic induction is used in alternators to generate current which alternates in direction (a.c.) and in dynamos to generate direct current (d.c.)	5b
13.4P Explain the action of the microphone in converting the pressure variations in sound waves into variations in current in electrical circuits, and the reverse effect as used in loudspeakers and headphones	5b
13.5 Explain how an alternating current in one circuit can induce a current in another circuit in a transformer	
13.6 Recall that a transformer can change the size of an alternating voltage	
13.7P Use the turns ratio equation for transformers to calculate either the missing voltage or the missing number of turns: <i>potential difference across primary coil</i> = <i>number of turns in primary coil</i> <i>potential difference across secondary coil</i> <i>number of turns in secondary coil</i> $\frac{V_p}{V_s} = \frac{N_p}{N_s}$	1a, 1c, 1d 2a 3a, 3b, 3c, 3d 5b
13.8 Explain why, in the national grid, electrical energy is transferred at high voltages from power stations, and then transferred at lower voltages in each locality for domestic uses as it improves the efficiency by reducing heat loss in transmission lines	
13.9 Explain where and why step-up and step-down transformers are used in the transmission of electricity in the national grid	

Students should:	Maths skills
13.10 Use the power equation (for transformers with 100% efficiency): potential difference across primary coil (volt, V) × current in primary coil (ampere, A) = potential difference across secondary coil (volt, V) × current in secondary coil (ampere, A) $V_p \times I_p = V_s \times I_s$	1a, 1c, 1d 2a 3a, 3b, 3c, 3d
13.11P Explain the advantages of power transmission in high-voltage cables, using the equations in 10.29, 10.31, 13.7P and 13.10	1a, 1c, 1d 2a 3a, 3b, 3c, 3d 5b

Use of mathematics

- Make calculations using ratios and proportional reasoning to convert units and to compute rates (1c, 3c).
- **Apply the equations linking the p.d.s and numbers of turns in the two coils of a transformer, to the currents and the power transfer involved, and relate these to the advantages of power transmission at high voltages (1c, 3b, 3c).**
- Make calculations of the energy changes associated with changes in a system, recalling or selecting the relevant equations for mechanical, electrical, and thermal processes; thereby express in quantitative form and on a common scale the overall redistribution of energy in the system (1a, 1c, 3c).

Suggested practicals

- Investigate factors affecting the generation of electric current by induction.